EAB 770 - Chapter 9

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1 Logistic Regression - Polytomous Response

1.1 Outline

- Ordinal Response
  - Proportional Odds Model
  - Multiple Qualitative Explanatory Variables
  - Partial Proportional Odds Model
- Nominal Response
  - Generalized Logits Model
  - Exact Methods: Generalized Logits Model

2 Ordinal Response Variables

2.1 Polytomous Variables?

Recall Chapter 8: dichotomous response variables can be analyzed using logistic regression.

> **Note**
>
> What if the response variable has more than two levels?

2.2 Polytomous variables

- Polytomous variables are usually described by the multinomial distribution

> **Tip**
>
> Unlike the binomial distribution where we only need one probability of success, more than one probability of success is needed to be defined in multinomial distributions.

2.3 Cumulative probabilities

Consider π_{hij} as the probability of success of the hth strata and i^{th} explanatory variables and the jth level of the response variable. We can define θ_{hij} such that:

$$\theta_{hiJ} = \sum_{j=1}^{J} \pi_{hij}$$

> **Note**
>
> If $J = 1$, then $\theta_{hi1} = \pi_{hi1}$. If $J = 2$, then $\theta_{hi2} = \pi_{hi1} + \pi_{hi2}$. θ_{hiJ} is the *cumulative probability* of the response levels from $j = 1$ to $j = J$.

> **Note**
>
> If there are $J + 1$ levels, then π_{J+1} = 1- _{hiJ}$.

2.4 Cumulative logits

Cumulative logits can be defined based on the cumulative probabilities θ_{hij}. For a specific level k out of K levels,

$$logit(\theta_{hik}) = \log\left(\frac{\theta_{hik}}{1-\theta_{hik}}\right) = \log\left(\frac{\sum_{j=1}^{k}\pi_{hij}}{1-\sum_{j=1}^{k}\pi_{hij}}\right) = \log\left(\frac{\sum_{j=1}^{k}\pi_{hij}}{\sum_{j=k+1}^{K}\pi_{hij}}\right)$$

2.5 Cumulative logits

Suppose we define a linear predictor η_{hij} such that

$$\eta_{hij} = logit(\theta_{hij}) = \alpha_j + \mathbf{x'_{hik}}$$

where $x'_{hi}\beta_k$ represents the matrix product of the explanatory/strata variables x_{hi} and the coefficients for each response level β_k.

2.6 Proportional Odds Assumption

Proportional Odds

The proportional odds assumption applies the following condition for a response variable with K total variable levels:

$$\boldsymbol{\beta_1 = \beta_2 = \beta_3 = \ldots = \beta_K = \beta}$$

Key assumption: the effect of the explanatory variables should not be disproportionate across all levels.

2.7 Proportional Odds

A consequence of the proportional odds assumption is that the difference between the levels of the explanatory variables are the same across different response levels.

Example

Consider comparing the mth and n^{th} level of the explanatory variable in the hth strata for the first and second level of the response variable.

$$\eta_{hm1} - \eta_{hn1} = (\mathbf{x'_{hm}} - \mathbf{x'_{hn}})\boldsymbol{\beta_1} = (\mathbf{x'_{hm}} - \mathbf{x'_{hn}})\boldsymbol{\beta}$$

$$\eta_{hm2} - \eta_{hn2} = (\mathbf{x'_{hm}} - \mathbf{x'_{hn}})\boldsymbol{\beta_2} = (\mathbf{x'_{hm}} - \mathbf{x'_{hn}})\boldsymbol{\beta}$$

These are the same across all levels of the response variable. The difference between η correspond to the log odds, hence the odds ratios between explanatory variables are the same across all response levels.

2.8 Proportional Odds

Another consequence of the proportional odds assumption is that the difference between the response levels does not depend on the strata or explanatory variables.

Example

Consider comparing the first and second level of the response variable.

$$\eta_{hm1} - \eta_{hm2} = \alpha_1 - \alpha_2 + (\mathbf{x'_{hm}} - \mathbf{x'_{hm}}\beta) = \alpha_1 - \alpha_2$$

> **Note**
>
> The proportional odds assumption can be tested using the Brant test. The null hypothesis of the test is that the proportional odds assumption holds. For the proportional odds assumption to hold, we must fail to reject the null hypothesis.

2.9 Solving for Individual Probabilities

Individual probabilities can be calculated by taking the differences between cumulative logits.

$$\pi_1 = \theta_1; \pi_2 = \theta_2 - \theta_1; \pi_k = \theta_k - \theta_{k-1}$$

2.10 Proportional Odds Model in SAS

- We can use the `LOGISTIC`, `GENMOD`, and `GLIMMIX` procedures to fit the proportional odds model.
- Dealing with ordinal variables means we need to specify order of the data.
- We can use the ESTIMATE and CONTRAST statements to determine and test the odds ratios between response levels, explanatory variable levels, and strata.
  - `ESTIMATE: EXP` option to get the estimated odds ratio, `ILINK` option to get the estimated cumulative probability θ.
  - CL option is used to get the confidence limits for odds ratio or estimated probabilities.

2.11 Proportional Odds Model in `R`

The `polr()` function in the `MASS` package in `R` performs the proportional odds model in `R`. Marginal/conditional means can still be calculated using `emmeans()` from the `emmeans` package.

> **Note**
>
> The `poTest()` function in `car` tests whether the proportional odds assumption holds.

> **Note**
>
> Because the `polr()` function uses a slightly different link between the cumulative logits, the interpretation for these numbers will be different from that of `SAS`. Calculating odds ratios will provide the odds ratio of recording higher level of outcomes between the two predictor levels.

2.12 Example

Male and female subjects received an active or placebo treatment for their arthritis pain, and the subsequent extent of improvement was recorded as marked, some, or none.

2.12.1 Questions

- Write the correct form of the linear predictor.
- Use the proportional odds model to determine whether there is an effect of sex and treatment.
- List the odds ratio estimates, 95% confidence intervals, and corresponding interpretations.
- Estimate the probability of marked improvement for females in the active group.

2.12.2 Data

Table 9.1 Arthrit

Sex	Treatment	Impr Marked
Female	Active	16
Female	Placebo	6
Male	Active	5
Male	Placebo	1

R Data Set

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
arthritis <- read.csv("arthritis_9.csv",stringsAsFactors = T)
glimpse(arthritis)
```

```
Rows: 84
Columns: 3
$ Treatment   <fct> Active, Active, Active, Active, Active, Active, Active, Ac~
$ Sex         <fct> Female, Female, Female, Female, Female, Female, Female, Fe~
$ Improvement <fct> Marked, Marked, Marked, Marked, Marked, Marked, Marked, Ma~
```

It is important to check if the factor levels follow the order we want.

```
levels(arthritis$Improvement)
```

```
[1] "Marked" "None"   "Some"
```

```
levels(arthritis$Treatment)
```

```
[1] "Active"  "Placebo"
```

```
levels(arthritis$Sex)
```

```
[1] "Female" "Male"
```

The code below fixes this.

```
arthritis <- arthritis |>
  mutate(Improvement = factor(Improvement, levels = c("None","Some", "Marked"),ordered=T),
         Treatment = factor(Treatment,levels=c("Placebo","Active")))
```

2.13 Example

Male and female subjects received an active or placebo treatment for their arthritis pain, and the subsequent extent of improvement was recorded as marked, some, or none.

2.13.1 SAS Data Set

```
data arthritis;
input sex $ treatment $ improve $ count @@;
datalines;
female active marked 16 female active some 5 female active none 6
female placebo marked 6 female placebo some 7 female placebo none 19
male active marked 5 male active some 2 male active none 7
male placebo marked 1 male placebo some 0 male placebo none 10
;
```

2.13.2 SAS Code

```
proc logistic data=arthritis order=data plots=effect(polybar x=treatment*sex); ①
freq count;
class treatment sex / param=ref;
model improve = sex treatment sex*treatment/ selection=forward include=2 scale=none aggregate
estimate "odds between marked vs. some+no improvement for female in placebo group" intercept
estimate "odds between marked+some vs. no improvement for female in placebo group" intercept
estimate "probability of marked improvement for female in the placebo group" intercept 1 sex
estimate "probability of marked improvement for female in the active trt group" intercept 1 s
run;
```

① The polybar plot creates stacked bar plots showing probability and cumulative probability plots for each level dictated by x.
② `selection=forward include=2` tests whether the interaction is needed to be included in the model.
③ The `ilink` option calculates the cumulative probability, while `exp` calculates odds ratios. The reason it returns the probability for marked is that the marked level was inputted as the lowest level.

2.13.3 R Code

```
library(MASS)
```

```
Attaching package: 'MASS'
```

```
The following object is masked from 'package:dplyr':

    select
```

```r
library(car)
```

```
Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

    recode

The following object is masked from 'package:purrr':

    some
```

```r
options(contrasts = c("contr.treatment", "contr.poly"))
mod_po <- polr(Improvement~Treatment+Sex,data=arthritis,Hess = TRUE)
summary(mod_po)
```

```
Call:
polr(formula = Improvement ~ Treatment + Sex, data = arthritis,
    Hess = TRUE)

Coefficients:
                Value Std. Error t value
TreatmentActive  1.797     0.4718   3.809
SexMale         -1.319     0.5381  -2.451

Intercepts:
            Value   Std. Error t value
None|Some    0.4941  0.3369     1.4665
Some|Marked  1.3485  0.3667     3.6768

Residual Deviance: 150.0294
AIC: 158.0294
```

```r
exp(coef(mod_po))
```

```
TreatmentActive         SexMale
      6.0330950       0.2674786
```

```
poTest(mod_po)
```

```
Tests for Proportional Odds
polr(formula = Improvement ~ Treatment + Sex, data = arthritis,
    Hess = TRUE)

                b[polr] b[>None] b[>Some] Chisquare df Pr(>Chisq)
Overall                                       1.83  2       0.40
TreatmentActive  1.79726  1.78168  1.81602      0.01  1       0.94
SexMale         -1.31872 -1.46865 -0.93005      1.58  1       0.21
```

```
Anova(mod_po)
```

```
Analysis of Deviance Table (Type II tests)

Response: Improvement
          LR Chisq Df Pr(>Chisq)
Treatment  16.0812  1  6.068e-05 ***
Sex         6.5766  1    0.01033 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
library(emmeans)
```

```
Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'
```

```
emm_cum <- emmeans(mod_po,~Treatment*Sex|cut,mode="cum.prob")
emm_cum
```

```
cut = None|Some:
 Treatment Sex    cumprob     SE  df asymp.LCL asymp.UCL
 Placebo   Female   0.621 0.0793 Inf    0.4657     0.776
 Active    Female   0.214 0.0658 Inf    0.0847     0.343
 Placebo   Male     0.860 0.0682 Inf    0.7260     0.993
 Active    Male     0.504 0.1180 Inf    0.2725     0.735

cut = Some|Marked:
```

```
 Treatment Sex    cumprob     SE  df asymp.LCL asymp.UCL
 Placebo   Female   0.794 0.0600 Inf    0.6763     0.912
 Active    Female   0.390 0.0870 Inf    0.2192     0.560
 Placebo   Male     0.935 0.0368 Inf    0.8629     1.007
 Active    Male     0.705 0.0999 Inf    0.5090     0.900

Confidence level used: 0.95
```

```r
emm_prob <- emmeans(mod_po,~Treatment*Sex|Improvement,mode="prob")
emm_prob
```

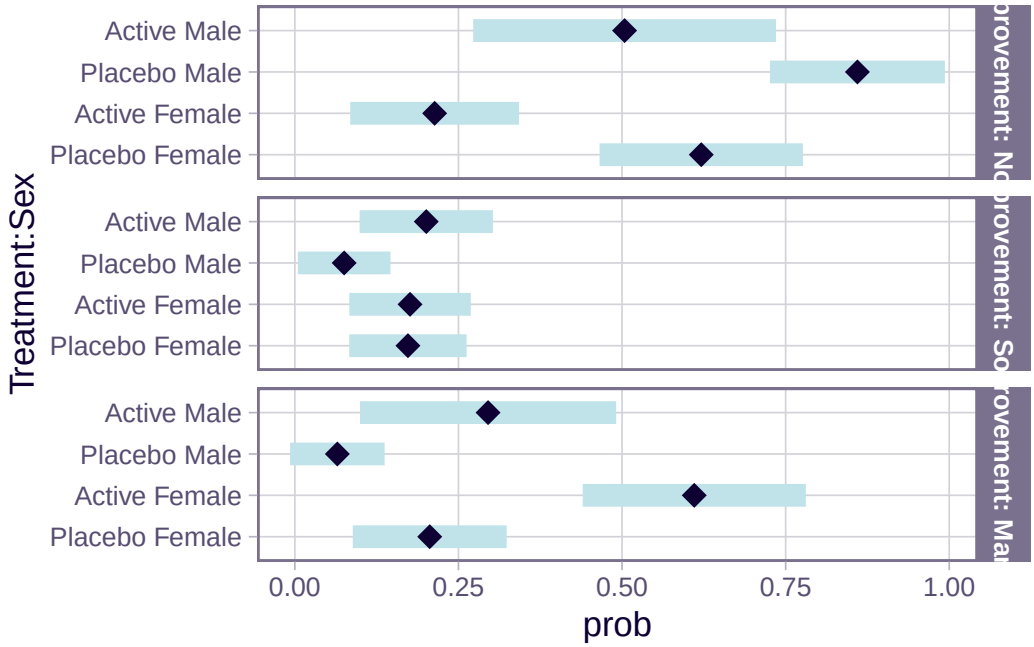
```
Improvement = None:
 Treatment Sex      prob     SE  df asymp.LCL asymp.UCL
 Placebo   Female 0.6211 0.0793 Inf   0.46567     0.776
 Active    Female 0.2136 0.0658 Inf   0.08466     0.343
 Placebo   Male   0.8597 0.0682 Inf   0.72603     0.993
 Active    Male   0.5039 0.1180 Inf   0.27255     0.735

Improvement = Some:
 Treatment Sex      prob     SE  df asymp.LCL asymp.UCL
 Placebo   Female 0.1728 0.0457 Inf   0.08315     0.262
 Active    Female 0.1760 0.0474 Inf   0.08320     0.269
 Placebo   Male   0.0754 0.0361 Inf   0.00454     0.146
 Active    Male   0.2008 0.0519 Inf   0.09904     0.303

Improvement = Marked:
 Treatment Sex      prob     SE  df asymp.LCL asymp.UCL
 Placebo   Female 0.2061 0.0600 Inf   0.08850     0.324
 Active    Female 0.6104 0.0870 Inf   0.43990     0.781
 Placebo   Male   0.0649 0.0368 Inf  -0.00724     0.137
 Active    Male   0.2953 0.0999 Inf   0.09954     0.491

Confidence level used: 0.95
```

```
emm_prob |> plot()
```



To calculate the odds ratio, we can use `contrast()` but we have to manually exponentiate the confidence limits.

```
emm_treatment <- emmeans(mod_po,~Treatment)
contrast(emm_treatment,method="revpairwise",infer=T) |>
  as.data.frame() |>
  mutate(OR = exp(estimate),
         OR.LCL = exp(asymp.LCL),
         OR.UCL = exp(asymp.UCL))
```

```
          contrast estimate        SE  df asymp.LCL asymp.UCL  z.ratio
1 Active - Placebo  1.79726 0.4717993 Inf 0.8725504   2.72197 3.809374
       p.value       OR   OR.LCL   OR.UCL
1 0.0001393191 6.033095 2.393006 15.21025
```

```
emm_sex <- emmeans(mod_po,~Sex)
contrast(emm_sex,method="pairwise",infer=T) |>
  as.data.frame() |>
  mutate(OR = exp(estimate),
```

```
          OR.LCL = exp(asymp.LCL),
          OR.UCL = exp(asymp.UCL))
```

```
       contrast estimate        SE  df asymp.LCL asymp.UCL  z.ratio    p.value
1 Female - Male 1.318716 0.5380997 Inf 0.2640599  2.373372 2.450691 0.01425824
        OR   OR.LCL   OR.UCL
1 3.738617 1.302206 10.73352
```

2.13.4 Answer

The linear predictor can be written as

$$\eta_k = \alpha_k + \beta_f x_f + \beta_a x_a$$

where $x_f = 1$ when the participant identifies as female, $x_a = 1$ when the participant is in the active treatment group. Using the residual chi-square test, the interaction term is not significant ($\chi^2 = 0.28, p = 0.60$).

Using the proportional odds model, we determine that there is a significant effect of sex ($\chi^2 = 6.2, p = 0.01$) and treatment ($\chi^2 = 14.4, p = 0.001$).

Females have 3.74 (95% CI: 1.33, 10.55) times the odds of marked improvement compared to males.

The odds of improvement after undergoing active treatment is 6.03 (95% CI: 2.39, 15.2) times that of the placebo group.

2.14 Exercise

Researchers collected information on subjects' exposure to general air pollution, exposure to pollution in their jobs, and whether they smoked. The response measured was chronic respiratory disease status. Subjects were assigned to one of four possible categories.

2.14.1 Questions

- Use the proportional odds main effects model to test the effect of air pollution, job exposure, and smoking status in the response.
- Estimate the odds ratio between Response I+II+III vs. IV for current vs. ex-smokers.
- Estimate the probability of developing a response level III for an ex-smoker in a high air pollution environment with job exposure.

2.14.2 Data Set

```
data respire;
input air $ exposure $ smoking $ level $ count @@;
datalines;
low no non 1 158 low no non 2 9
low no ex 1 167 low no ex 2 19
low no cur 1 307 low no cur 2 102
low yes non 1 26 low yes non 2 5
low yes ex 1 38 low yes ex 2 12
low yes cur 1 94 low yes cur 2 48
high no non 1 94 high no non 2 7
high no ex 1 67 high no ex 2 8
high no cur 1 184 high no cur 2 65
high yes non 1 32 high yes non 2 3
high yes ex 1 39 high yes ex 2 11
high yes cur 1 77 high yes cur 2 48
low no non 3 5 low no non 4 0
low no ex 3 5 low no ex 4 3
low no cur 3 83 low no cur 4 68
low yes non 3 5 low yes non 4 1
low yes ex 3 4 low yes ex 4 4
low yes cur 3 46 low yes cur 4 60
high no non 3 5 high no non 4 1
high no ex 3 4 high no ex 4 3
high no cur 3 33 high no cur 4 36
high yes non 3 6 high yes non 4 1
high yes ex 3 4 high yes ex 4 2
high yes cur 3 39 high yes cur 4 51
;
```

A `csv` file is uploaded on Canvas as `air_pollution.csv`.

```
library(tidyverse)
air_pollution <- read.csv("air_pollution.csv",stringsAsFactors = TRUE)
air_pollution <- air_pollution |>
  mutate(Level = factor(Level, ordered=T))

glimpse(air_pollution)
```

```
Rows: 2,089
```

```
Columns: 4
$ Air      <fct> high, high, high, high, high, high, high, high, high, high, h~
$ Exposure <fct> yes, yes, yes, yes, yes, yes, yes, yes, yes, yes, yes, yes, y~
$ Smoking  <fct> cur, cur, cur, cur, cur, cur, cur, cur, cur, cur, cur, cur, c~
$ Level    <ord> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4~
```

2.15 Exercise

Researchers collected information on subjects' exposure to general air pollution, exposure to pollution in their jobs, and whether they smoked. The response measured was chronic respiratory disease status. Subjects were assigned to one of four possible categories.

2.15.1 SAS Code

```
proc logistic data=respire order=data plots=effect(polybar x=air*exposure*smoking);;
freq count;
class air exposure smoking / param=reference;
model level = air exposure smoking air*exposure exposure*smoking air*smoking air*exposure*sm
estimate "odds between I+II+III vs. IV for current vs. ex smokers" smoking 1 -1 /exp cl categ
estimate "probability of at most III level for ex smoker, high air pollution, job exposure" i
estimate "probability of at most II level for ex smoker, high air pollution, job exposure" in
run;
```

2.15.2 R Code

```
library(MASS)
library(car)
options(contrasts = c("contr.treatment", "contr.poly"))
mod_po <- polr(Level~Air + Exposure + Smoking, data=air_pollution,Hess=TRUE)

summary(mod_po)
```

```
Call:
polr(formula = Level ~ Air + Exposure + Smoking, data = air_pollution,
    Hess = TRUE)

Coefficients:
              Value Std. Error  t value
Airlow      0.03929    0.09384   0.4187
```

```
Exposureyes  0.86476    0.09576   9.0302
Smokingex   -1.45268    0.13725 -10.5842
Smokingnon  -1.85271    0.16481 -11.2416

Intercepts:
    Value    Std. Error t value
1|2   0.2750   0.0880     3.1256
2|3   1.1562   0.0924    12.5133
3|4   2.0804   0.1044    19.9318

Residual Deviance: 4178.511
AIC: 4192.511
```

```
Anova(mod_po)
```

```
Analysis of Deviance Table (Type II tests)

Response: Level
         LR Chisq Df Pr(>Chisq)
Air         0.175  1     0.6753
Exposure   81.308  1     <2e-16 ***
Smoking   255.541  2     <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
poTest(mod_po)
```

```
Tests for Proportional Odds
polr(formula = Level ~ Air + Exposure + Smoking, data = air_pollution,
    Hess = TRUE)

             b[polr]    b[>1]    b[>2]    b[>3] Chisquare df Pr(>Chisq)
Overall                                            10.66  8      0.222
Airlow       0.03929  0.00592  0.10350  0.02952      1.39  2      0.499
Exposureyes  0.86476  0.84954  0.85999  0.87715      0.04  2      0.981
Smokingex   -1.45268 -1.39976 -1.71927 -1.76946      3.61  2      0.165
Smokingnon  -1.85271 -1.83705 -1.75516 -3.02210      5.53  2      0.063 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
library(emmeans)
emm_cum <- emmeans(mod_po,~Air+Exposure+Smoking|cut,mode="cum.prob")
emm_cum
```

```
cut = 1|2:
 Air  Exposure Smoking cumprob      SE  df asymp.LCL asymp.UCL
 high no       cur       0.568 0.02160 Inf     0.526     0.611
 low  no       cur       0.559 0.01790 Inf     0.524     0.594
 high yes      cur       0.357 0.02200 Inf     0.313     0.400
 low  yes      cur       0.348 0.02140 Inf     0.306     0.390
 high no       ex        0.849 0.01880 Inf     0.812     0.886
 low  no       ex        0.844 0.01780 Inf     0.809     0.879
 high yes      ex        0.703 0.03090 Inf     0.643     0.764
 low  yes      ex        0.695 0.03060 Inf     0.635     0.755
 high no       non       0.894 0.01610 Inf     0.862     0.925
 low  no       non       0.890 0.01600 Inf     0.858     0.921
 high yes      non       0.780 0.03000 Inf     0.721     0.838
 low  yes      non       0.773 0.03070 Inf     0.713     0.833

cut = 2|3:
 Air  Exposure Smoking cumprob      SE  df asymp.LCL asymp.UCL
 high no       cur       0.761 0.01680 Inf     0.728     0.794
 low  no       cur       0.753 0.01440 Inf     0.725     0.782
 high yes      cur       0.572 0.02350 Inf     0.526     0.618
 low  yes      cur       0.563 0.02310 Inf     0.517     0.608
 high no       ex        0.931 0.00978 Inf     0.912     0.951
 low  no       ex        0.929 0.00939 Inf     0.910     0.947
 high yes      ex        0.851 0.01920 Inf     0.814     0.889
 low  yes      ex        0.846 0.01930 Inf     0.808     0.884
 high no       non       0.953 0.00784 Inf     0.938     0.968
 low  no       non       0.951 0.00783 Inf     0.936     0.967
 high yes      non       0.895 0.01670 Inf     0.862     0.928
 low  yes      non       0.891 0.01720 Inf     0.858     0.925

cut = 3|4:
 Air  Exposure Smoking cumprob      SE  df asymp.LCL asymp.UCL
 high no       cur       0.889 0.01030 Inf     0.869     0.909
 low  no       cur       0.885 0.00923 Inf     0.867     0.903
 high yes      cur       0.771 0.01830 Inf     0.735     0.807
 low  yes      cur       0.764 0.01830 Inf     0.728     0.800
 high no       ex        0.972 0.00450 Inf     0.963     0.980
 low  no       ex        0.971 0.00436 Inf     0.962     0.979
```

```
 high yes      ex        0.935 0.00965 Inf     0.916     0.954
 low  yes      ex        0.933 0.00976 Inf     0.914     0.952
 high no       non       0.981 0.00346 Inf     0.974     0.988
 low  no       non       0.980 0.00347 Inf     0.973     0.987
 high yes      non       0.956 0.00782 Inf     0.940     0.971
 low  yes      non       0.954 0.00809 Inf     0.938     0.970

Confidence level used: 0.95
```

```
emm_prob <- emmeans(mod_po,~Air+Exposure+Smoking|Level,mode="prob")
emm_prob
```

```
Level = 1:
 Air  Exposure Smoking   prob      SE  df asymp.LCL asymp.UCL
 high no       cur     0.5683 0.02160 Inf    0.5260    0.6106
 low  no       cur     0.5587 0.01790 Inf    0.5236    0.5937
 high yes      cur     0.3567 0.02200 Inf    0.3135    0.3999
 low  yes      cur     0.3477 0.02140 Inf    0.3057    0.3898
 high no       ex      0.8491 0.01880 Inf    0.8123    0.8860
 low  no       ex      0.8440 0.01780 Inf    0.8091    0.8790
 high yes      ex      0.7033 0.03090 Inf    0.6428    0.7638
 low  yes      ex      0.6950 0.03060 Inf    0.6350    0.7551
 high no       non     0.8936 0.01610 Inf    0.8620    0.9251
 low  no       non     0.8898 0.01600 Inf    0.8585    0.9211
 high yes      non     0.7795 0.03000 Inf    0.7207    0.8383
 low  yes      non     0.7727 0.03070 Inf    0.7125    0.8329

Level = 2:
 Air  Exposure Smoking   prob      SE  df asymp.LCL asymp.UCL
 high no       cur     0.1923 0.01060 Inf    0.1716    0.2130
 low  no       cur     0.1948 0.01020 Inf    0.1749    0.2147
 high yes      cur     0.2157 0.01070 Inf    0.1947    0.2366
 low  yes      cur     0.2150 0.01070 Inf    0.1940    0.2360
 high no       ex      0.0823 0.00987 Inf    0.0630    0.1017
 low  no       ex      0.0849 0.00940 Inf    0.0664    0.1033
 high yes      ex      0.1479 0.01370 Inf    0.1210    0.1749
 low  yes      ex      0.1512 0.01360 Inf    0.1245    0.1778
 high no       non     0.0594 0.00875 Inf    0.0423    0.0766
 low  no       non     0.0614 0.00868 Inf    0.0444    0.0784
 high yes      non     0.1156 0.01450 Inf    0.0872    0.1439
 low  yes      non     0.1187 0.01470 Inf    0.0899    0.1475
```

```
Level = 3:
 Air  Exposure Smoking   prob      SE  df asymp.LCL asymp.UCL
 high no       cur     0.1283 0.00992 Inf    0.1089    0.1478
 low  no       cur     0.1316 0.00919 Inf    0.1136    0.1496
 high yes      cur     0.1989 0.01280 Inf    0.1738    0.2240
 low  yes      cur     0.2016 0.01280 Inf    0.1764    0.2267
 high no       ex      0.0402 0.00579 Inf    0.0288    0.0515
 low  no       ex      0.0416 0.00559 Inf    0.0307    0.0526
 high yes      ex      0.0839 0.01070 Inf    0.0629    0.1050
 low  yes      ex      0.0865 0.01080 Inf    0.0654    0.1077
 high no       non     0.0278 0.00468 Inf    0.0186    0.0370
 low  no       non     0.0289 0.00468 Inf    0.0197    0.0380
 high yes      non     0.0604 0.00957 Inf    0.0417    0.0792
 low  yes      non     0.0625 0.00985 Inf    0.0432    0.0818

Level = 4:
 Air  Exposure Smoking   prob      SE  df asymp.LCL asymp.UCL
 high no       cur     0.1110 0.01030 Inf    0.0908    0.1312
 low  no       cur     0.1150 0.00923 Inf    0.0969    0.1330
 high yes      cur     0.2287 0.01830 Inf    0.1928    0.2646
 low  yes      cur     0.2357 0.01830 Inf    0.1998    0.2716
 high no       ex      0.0284 0.00450 Inf    0.0196    0.0372
 low  no       ex      0.0295 0.00436 Inf    0.0209    0.0380
 high yes      ex      0.0649 0.00965 Inf    0.0459    0.0838
 low  yes      ex      0.0673 0.00976 Inf    0.0482    0.0864
 high no       non     0.0192 0.00346 Inf    0.0124    0.0260
 low  no       non     0.0200 0.00347 Inf    0.0132    0.0268
 high yes      non     0.0444 0.00782 Inf    0.0291    0.0598
 low  yes      non     0.0461 0.00809 Inf    0.0303    0.0620

Confidence level used: 0.95
```

To calculate the odds ratio, we can use `contrast()` but we have to manually exponentiate the confidence limits.

```
emm_Air <- emmeans(mod_po,~Air)
contrast(emm_Air,method="revpairwise",infer=T) |>
  as.data.frame() |>
  mutate(OR = exp(estimate),
         OR.LCL = exp(asymp.LCL),
         OR.UCL = exp(asymp.UCL))
```

```
    contrast  estimate         SE  df  asymp.LCL asymp.UCL   z.ratio   p.value
```

```
1 low - high 0.0392934 0.09384313 Inf -0.1446358 0.2232226 0.4187136 0.6754255
        OR    OR.LCL   OR.UCL
1 1.040076 0.8653374 1.250099
```

```
emm_Ex <- emmeans(mod_po,~Exposure)
contrast(emm_Ex,method="revpairwise",infer=T) |>
  as.data.frame() |>
  mutate(OR = exp(estimate),
         OR.LCL = exp(asymp.LCL),
         OR.UCL = exp(asymp.UCL))
```

```
  contrast  estimate         SE  df asymp.LCL asymp.UCL  z.ratio      p.value
1 yes - no 0.8647616 0.09576341 Inf 0.6770688  1.052454 9.030189 1.713763e-19
       OR OR.LCL   OR.UCL
1 2.37444 1.9681 2.864674
```

```
emm_Smoking <- emmeans(mod_po,~Smoking)
contrast(emm_Smoking,method="pairwise",infer=T,adjust="tukey") |>
  as.data.frame() |>
  mutate(OR = exp(estimate),
         OR.LCL = exp(asymp.LCL),
         OR.UCL = exp(asymp.UCL))
```

```
   contrast  estimate         SE  df   asymp.LCL asymp.UCL   z.ratio
1  cur - ex 1.4526828 0.1372496 Inf  1.13101079 1.7743549 10.584238
2 cur - non 1.8527107 0.1648078 Inf  1.46645064 2.2389708 11.241647
3  ex - non 0.4000279 0.2007087 Inf -0.07037313 0.8704288  1.993077
       p.value       OR   OR.LCL   OR.UCL
1 2.786660e-14 4.274567 3.098787 5.896476
2 5.329071e-15 6.377082 4.333826 9.383668
3 1.138789e-01 1.491866 0.932046 2.387935
```

2.15.3 Answer

There is no evidence of an air pollution effect ($\chi^2 = 0.18, p = 0.68$). There is evidence of a strong association between exposure ($\chi^2 = 82.1, p =< 0.001$) and smoking history ($\chi^2 = 209.9, p < 0.001$).

The odds of being assigned to Response Level III and lower for current smokers is 0.23 times the odds for ex-smokers. The odds of being assigned to Response Level 4 for current smokers is 4.27 times the odds for ex-smokers.

The probability of receiving a response level III and below for an ex smoker, high air pollution, and with job exposure is 0.9351. The probability of receiving a response level II and below for an ex smoker, high air pollution, and with job exposure is 0.8512. Hence, the estimated probability of receiving a response level III is 0.0839 .

2.16 Partial Proportional Odds Model

Sometimes, the $\beta_k = \beta$ assumption does not hold for all explanatory variables, especially if some levels disproportionately affect the odds of success.

> **Tip**
>
> This assumption can be tested using the score test for the proportional odds assumption provided by the `LOGISTIC` procedure.

To address this, we specify unequal slopes for the explanatory variables that violate the proportional odds assumption.

$$logit(\theta_{hik}) = \alpha_k + x'_{hi}\beta_k + x'_{h'i'}\gamma$$

where x'_{hi} includes the explanatory variables with unequal slopes β_k and $x'_{h'i'}$ includes the explanatory variables with equal slopes γ. This model is called the **partial proportional odds model**.

2.17 Partial Proportional Odds Model in SAS

- If the score test for the proportional odds assumption is significant at the 0.10 level, we might need to use a partial proportional odds model.
- We can specify the model to have unequal slopes using the `UNEQUALSLOPES` option in the MODEL statement. `UNEQUALSLOPES=X` specifies that only the explanatory variable X has unequal slopes
- Testing for unequal slopes can be done using the `TEST` statement
  - `LABEL`: TEST statements effectively labels the test.
  - If the test yields a significant result, there is sufficient evidence for unequal slopes for that explanatory variable.
  - If the test is not significant, we can assume the proportional odds model holds.

2.18 Partial Proportional Odds Model in R

If the score test for the proportional odds assumption led to the rejection of the null hypothesis at the 0.10 level, we can use a partial proportional odds model.

> **Note**
>
> The `vglm` package in the `VGAM` package in `R` can be used to perform partial proportional odds model. Suppose we have two predictors, `x1` and `x2`, with `x2` not satisfying the proportional odds model.

```
mod_vglm <- vglm(outcome~x1 + x2,
                 data= sample_data,
                 family=cumulative(parallel=FALSE~x2))
```

> It can also be used to estimate proportional odds models. However, `vglm()` does not work with `emmeans` as of September 2026.

2.19 Example

The table contains data from a study by a bicycling clothing manufacturer who wanted to assess their test glove. It was designed to combat the hand problems experienced by cyclists dealing with carpal tunnel syndrome. Cyclists wore either the company's standard gel glove or the new test glove for a week's worth of their standard rides. Then they reported whether they experienced major, moderate, or no relief from their usual symptoms (numbness and nerve pain) on the bike.

2.19.1 Questions

- Assume an unequal slopes, main effect model and test for the equality of slopes for glove and gender.
- Perform logistic regression and interpret the results.

2.19.2 Data

```
data wrist;
input glove $ gender $ relief $ count @@;
datalines;
test female major 12 test female moderate 8 test female none 5
test male major 8 test male moderate 14 test male none 15
```

```
gel female major 15 gel female moderate 5 gel female none 9
gel male major 8 gel male moderate 4 gel male none 20
;
```

2.19.3 R Data

```
library(tidyverse)
cycling <- expand.grid(Relief = c("Major","Moderate","None"),
                       Gender = c("Female","Male"),
                       Type = c("Test","Gel"))

cycling$count <- c(12,8,5,8,14,15,15,5,9,8,4,20)
cycling <- cycling |>
  mutate(Relief = ordered(Relief, levels = c("Major","Moderate","None"))) |>
  uncount(weights=count)
```

2.20 Example

The table contains data from a study by a bicycling clothing manufacturer who wanted to assess their test glove. It was designed to combat the hand problems experienced by cyclists dealing with carpal tunnel syndrome. Cyclists wore either the company's standard gel glove or the new test glove for a week's worth of their standard rides. Then they reported whether they experienced major, moderate, or no relief from their usual symptoms (numbness and nerve pain) on the bike.

2.20.1 SAS Code

```
proc logistic order=data data=wrist;
freq count;
class glove gender / param=reference order=data;
model relief= glove gender / scale=none aggregate unequalslopes;          ①
pogender: test genderfemale_major=genderfemale_moderate;                  ②
poglove: test glovetest_major=glovetest_moderate;                         ③
run;

proc logistic order=data data=wrist;
freq count;
class glove gender / param=reference order=data;
```

```
model relief= glove gender / scale=none aggregate unequalslopes=glove;      ④
run;
```

① `unequal slopes` tells SAS to provide unequal slopes for each explanatory level at all response levels.
② `pogender` is the label for testing whether the slopes should be the same for different response levels for the gender explanatory variable
③ `poglove` is the label for testing whether the slopes should be the same for different response levels for the glove explanatory variable
④ Only assumes unequal slopes for glove, but proportional odds assumption holds for gender.

2.20.2 R Code

```
library(MASS)
library(ordinal)
```

```
Warning: package 'ordinal' was built under R version 4.6.1
```

```
Attaching package: 'ordinal'
```

```
The following object is masked from 'package:dplyr':

    slice
```

```
library(car)
mod_po <- polr(Relief~Gender+Type,data=cycling)
poTest(mod_po)
```

```
Tests for Proportional Odds
polr(formula = Relief ~ Gender + Type, data = cycling)

           b[polr] b[>Major] b[>Moderate] Chisquare df Pr(>Chisq)
Overall                                        6.85  2     0.0326 *
GenderMale  1.1766    1.1873       1.1747      0.00  1     0.9724
TypeGel     0.3504   -0.1683       0.7811      6.75  1     0.0094 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The proportional odds test shows that there is evidence of unequal slopes for type, but not for gender.

```
library(VGAM)
```

```
Warning: package 'VGAM' was built under R version 4.6.1
```

```
Loading required package: stats4
```

```
Loading required package: splines
```

```
Attaching package: 'VGAM'
```

```
The following objects are masked from 'package:ordinal':

    dgumbel, dlgamma, pgumbel, plgamma, qgumbel, rgumbel, wine
```

```
library(car)
modv<- vglm(Relief~Gender+Type,
    family = cumulative(parallel=FALSE~ Type),
    data=cycling)

summary(modv)
```

```
Call:
vglm(formula = Relief ~ Gender + Type, family = cumulative(parallel = FALSE ~
    Type), data = cycling)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept):1  -0.0826     0.3379  -0.244 0.806893
(Intercept):2   1.5170     0.3677   4.125  3.7e-05 ***
GenderMale     -1.1819     0.3548  -3.331 0.000866 ***
TypeGel:1       0.1485     0.3926   0.378 0.705225
TypeGel:2      -0.7952     0.3894  -2.042 0.041167 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: logitlink(P[Y<=1]), logitlink(P[Y<=2])
```

```
Residual deviance: 247.0353 on 241 degrees of freedom

Log-likelihood: -123.5177 on 241 degrees of freedom

Number of Fisher scoring iterations: 4


Exponentiated coefficients:
GenderMale  TypeGel:1  TypeGel:2
 0.3067019  1.1600929  0.4515106
```

```
Anova(modv)
```

```
Analysis of Deviance Table (Type II tests)

Response: Relief
       Df   Chisq Pr(>Chisq)
Gender  1 11.0946  0.0008658 ***
Type    2  7.7644  0.0206052 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Let's calculate the probabilities and odds ratios.

```
# Odds Ratio

exp(coef(modv))
```

```
(Intercept):1 (Intercept):2    GenderMale     TypeGel:1     TypeGel:2
    0.9207215     4.5584608     0.3067019     1.1600929     0.4515106
```

```
exp(confint(modv))
```

```
                  2.5 %    97.5 %
(Intercept):1 0.4747861 1.7854949
(Intercept):2 2.2172362 9.3718316
GenderMale    0.1529985 0.6148169
TypeGel:1     0.5374483 2.5040837
TypeGel:2     0.2104667 0.9686180
```

```
# For reverse pairwise order: (note that the limits are reversed.)
exp(-coef(modv))
```

```
(Intercept):1 (Intercept):2    GenderMale     TypeGel:1     TypeGel:2
    1.0861047     0.2193723     3.2604949     0.8619999     2.2147872
```

```
exp(-confint(modv))
```

```
                  2.5 %    97.5 %
(Intercept):1 2.1062117 0.5600688
(Intercept):2 0.4510119 0.1067027
GenderMale    6.5360121 1.6265005
TypeGel:1     1.8606441 0.3993477
TypeGel:2     4.7513445 1.0323988
```

2.20.3 Answer

Using a partial proportional odds model, there is a significant effect of the type of glove and gender. the odds of experiencing major relief for females is 3.26 (95% CI: 1.63, 6.54) times that of males. The odds of major relief when using the test glove is 1.16 (0.54, 2.50) times that of gel. Additionally, the odds of moderate relief when using the test glove is 2.22 (1.03, 4.75) times that of gel.

2.21 Exercise

Consider the arthritis example discussed earlier. Test the hypothesis of equal slopes for both sex and treatment.

2.22 Exercise

Consider the arthritis example discussed earlier. Test the hypothesis of equal slopes for both sex and treatment.

2.22.1 SAS Code

```
proc logistic data=arthritis order=data plots=effect(polybar x=treatment*sex);;
freq count;
class treatment sex / param=reference;
model improve = sex treatment / unequalslopes scale=none aggregate;
equal_sex: test sexfemale_marked=sexfemale_some;
equal_treatment: test treatmentactive_marked=treatmentactive_some;
run;
```

2.22.2 R Code

```
mod_po <- polr(Improvement~Sex+Treatment,data=arthritis)
poTest(mod_po)
```

```
Tests for Proportional Odds
polr(formula = Improvement ~ Sex + Treatment, data = arthritis)

                 b[polr] b[>None] b[>Some] Chisquare df Pr(>Chisq)
Overall                                         1.83  2       0.40
SexMale         -1.31872 -1.46865 -0.93005      1.58  1       0.21
TreatmentActive  1.79726  1.78168  1.81602      0.01  1       0.94
```

2.22.3 Answer

We suppose the slopes are unequal, then test whether there is evidence against the null hypothesis of equal slopes using `test`.

`SAS`: There is insufficient evidence to claim that the β for sex ($\chi^2 = 1.35, p = 0.25$) and treatment ($\chi^2 = 0.03, p = 0.86$) are different across the different outcome levels. `R`: There is insufficient evidence to claim that the β for sex ($\chi^2 = 1.58, p = 0.21$) and treatment ($\chi^2 = 0.01, p = 0.94$) are different across the different outcome levels.

3 Nominal Response Variables

3.1 Generalized Logits Model

When the response variables are nominal, the concept of cumulative probabilities is less applicable. The **generalized logits model** is used for nominal polytomous response variables.

Generalized Logit Model

The logit function is the logarithm of the ratio of the probability of the level of interest, π_j and the *reference* category π_r.

$$logit\theta_j = \log\left(\frac{\pi_j}{\pi_r}\right) = \alpha_j + \sum_g \beta_{jg} x_g$$

where g comprises of ALL explanatory variables

Reference level

For the reference level, $\alpha_r = 0, \beta_{rg} = 0$.

3.2 Generalized Logits Model

The resulting expression for π_j can be written as:

$$\pi_j = \frac{\exp[\alpha_j + \sum_g \beta_{jg} x_g]}{\sum_{k=1}^{r} \exp[\alpha_k + \sum_g \beta_{kg} x_g]}$$

with the constraint $\alpha_r = 0, \beta_{rg} = 0$.

3.3 Generalized Logits Model: SAS

Implementing generalized logit models in SAS

The `LOGISTIC` procedure will model the logit models for all levels simultaneously. If there are r levels of the response variable, there will be $(r-1)$ degrees of freedom for a single logit model.

Link function

SAS does not know whether your response variable is ordinal or nominal. The generalized logits model can be specified using `link=glogit` in the `model` statement.

Syntax

- Unless specified, the last variable level based on the specified order in the `class` statement is set as the reference level.
- Syntax is the same as the PO model, but the assumptions are different.

3.4 Generalized Logits Model: `R`

The `vglm()` function in the `VGAM` package performs the generalized logits model using the `multinomial()` function to define the family of models.

```
modv <- vglm(Outcome~x1 + x2,
             data=sample_data,
             family=multinomial(parallel=FALSE, refLevel=1)) #reflevel=1 just specifies the r
```

3.5 Example

Schoolchildren in experimental learning settings were surveyed to determine which learning style they preferred. Investigators were interested in whether response was associated with school and their program, which was either a standard school day or also included after-hours care.

3.5.1 Question

Use logistic regression to test for program and school effects and calculate the relevant odds ratios.

3.5.2 SAS Data Set

```
data school;
input school program $ style $ count @@;
datalines;
1 regular self 10 1 regular team 17 1 regular class 26
1 after self 5 1 after team 12 1 after class 50
2 regular self 21 2 regular team 17 2 regular class 26
2 after self 16 2 after team 12 2 after class 36
3 regular self 15 3 regular team 15 3 regular class 16
3 after self 12 3 after team 12 3 after class 20
```

```
;

proc freq data=school order=data;
weight count;
tables school*program*style/ nocol norow nopct;
run;
```

3.5.3 R Data Set

```
library(tidyverse)
school <- expand.grid(Program = c("Regular","After"),
                      Style = c("Self","Team","Class"),
                      School = c("1","2","3"))

school$Count <- c(10,5,17,12,26,50,21,16,17,12,26,36,15,12,15,12,16,20)

school <- school |>
  uncount(weights = Count)
```

3.6 Example

Schoolchildren in experimental learning settings were surveyed to determine which learning style they preferred. Investigators were interested in whether response was associated with school and their program, which was either a standard school day or also included after-hours care. Use logistic regression to test for program and school effects and calculate the relevant odds ratios.

3.6.1 SAS Code

```
proc logistic data=school order=data;
freq count;
class school (ref=first) program (ref=last) / param=ref;
model style= program school school*program / link=glogit selection=forward include=2 scale=n
run;
```

3.6.2 R Code

```r
modv <- vglm(Style~School + Program,
             data=school,
             family=multinomial(parallel=FALSE, refLevel=3))

summary(modv)
```

```
Call:
vglm(formula = Style ~ School + Program, family = multinomial(parallel = FALSE,
    refLevel = 3), data = school)

Coefficients:
               Estimate Std. Error z value Pr(>|z|)
(Intercept):1   -1.2233     0.3154  -3.879 0.000105 ***
(Intercept):2   -0.5662     0.2586  -2.189 0.028593 *
School2:1        1.0828     0.3539   3.059 0.002218 **
School2:2        0.1801     0.3172   0.568 0.570163
School3:1        1.3147     0.3839   3.424 0.000616 ***
School3:2        0.6556     0.3395   1.931 0.053455 .
ProgramAfter:1  -0.7474     0.2820  -2.651 0.008028 **
ProgramAfter:2  -0.7426     0.2706  -2.745 0.006057 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,1]/mu[,3]), log(mu[,2]/mu[,3])

Residual deviance: 666.9339 on 668 degrees of freedom

Log-likelihood: -333.4669 on 668 degrees of freedom

Number of Fisher scoring iterations: 4


Reference group is level  3  of the response
```

```r
library(car)
Anova(modv)
```

```
Analysis of Deviance Table (Type II tests)

Response: Style
```

```
        Df  Chisq Pr(>Chisq)
School   4 14.843   0.005039 **
Program  2 10.916   0.004262 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Odds Ratio
exp(coef(modv))
```

```
 (Intercept):1  (Intercept):2      School2:1      School2:2      School3:1
     0.2942532      0.5676812      2.9530080      1.1973459      3.7236794
     School3:2 ProgramAfter:1 ProgramAfter:2
     1.9262906      0.4735774      0.4758599
```

```r
exp(confint(modv))
```

```
                    2.5 %    97.5 %
(Intercept):1  0.1585883 0.5459732
(Intercept):2  0.3419345 0.9424668
School2:1      1.4757049 5.9092141
School2:2      0.6430168 2.2295485
School3:1      1.7545863 7.9025968
School3:2      0.9902930 3.7469675
ProgramAfter:1 0.2725131 0.8229898
ProgramAfter:2 0.2800064 0.8087053
```

3.6.3 Answer

There is sufficient evidence of a program ($\chi^2 = 10.9, p = 0.004$) and school ($\chi^2 = 14.8, p = 0.005$).

The odds of preferring self-directed learning for those assigned to after-hours care is 0.47 (0.27-0.82) times that of those who assigned to regular class hours. The odds of students from school 2 preferring self-directed learning is 2.95 (1.48-5.91) times that of students from school 1. Similarly, the odds of students from school 3 preferring self-directed learning is 3.72 (1.76-7.90) times that of students from school 1.

The odds of preferring team-based learning for those assigned to after-hours care is 0.48 (0.27-0.82) times that of those who assigned to regular class hours. The odds of students from school 2 preferring self-directed learning is 1.20 (0.64-2.23) times that of students from school 1. Similarly, the odds of students from school 3 preferring self-directed learning is 1.93 (0.99-3.75) times that of students from school 1.

3.7 Exercise

The data set "firecracker" shows data of injuries due to firecrackers from three community health centers in rural Philippines. The age groups were coded as A: <25 years of age, B: 25 years of age. Use a generalized logit main effect model to determine whether there is a significant effect of age and clinic in the occurrence of face and finger injury compared to other parts of the body.

3.7.1 Question

Use the estimate statement to test and estimate the odds ratio between clinics 1 and 2 for fingers vs. other injury and face vs. other injury.

3.7.2 Data

```
data firecracker;
input clinic $ agegroup $ injury $ count;
datalines;
1 A Fingers 25
1 A Face 18
1 A Other 5
1 B Fingers 13
1 B Face 5
1 B Other 3
2 A Fingers 42
2 A Face 7
2 A Other 12
2 B Fingers 23
2 B Face 5
2 B Other 5
3 A Fingers 10
3 A Face 4
3 A Other 4
3 B Fingers 9
3 B Face 7
3 B Other 5
;
run;
```

3.7.3 R Data

```
firecracker <- expand.grid(Injury=c("Fingers","Face","Other"),
                           Agegroup=c("A","B"),
                           Clinic = c("1","2","3"))

firecracker$Count <- c(25,18,5,13,5,3,42,7,12,23,5,5,10,4,4,9,7,5)
firecracker <- firecracker |>
  uncount(weights=Count)
```

3.8 Exercise

The data set "firecracker" shows data of injuries due to firecrackers from three community health centers in rural Philippines. The age groups were coded as A: <25 years of age, B: 25 years of age. Use a generalized logit main effect model to determine whether there is a significant effect of age and clinic in the occurrence of face and finger injury compared to other parts of the body.

3.8.1 SAS Code

```
proc logistic data=firecracker order=data;
freq count;
class clinic agegroup;
model injury= clinic agegroup clinic*agegroup/link=glogit selection=forward include=2 scale=
estimate "clinic 1 vs clinic 2 - face" clinic 1 -1/ exp cl category="Face";
estimate "clinic 1 vs clinic 2 - fingers" clinic 1 -1/ exp cl category="Fingers";
run;
```

3.8.2 R Code

```
modv <- vglm(Injury~Clinic+Agegroup,
             data=firecracker,
             family=multinomial(parallel=FALSE, refLevel=3))

summary(modv)
```

```
Call:
vglm(formula = Injury ~ Clinic + Agegroup, family = multinomial(parallel = FALSE,
```

```
    refLevel = 3), data = firecracker)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept):1  1.54406    0.40779   3.786 0.000153 ***
(Intercept):2  1.05057    0.43512   2.414 0.015759 *
Clinic2:1     -0.21911    0.47524  -0.461 0.644763
Clinic2:2     -1.40520    0.55777  -2.519 0.011758 *
Clinic3:1     -0.82186    0.56949  -1.443 0.148976
Clinic3:2     -0.85967    0.61912  -1.389 0.164975
AgegroupB:1    0.04654    0.40891   0.114 0.909377
AgegroupB:2    0.01831    0.48319   0.038 0.969767
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,1]/mu[,3]), log(mu[,2]/mu[,3])

Residual deviance: 367.3984 on 396 degrees of freedom

Log-likelihood: -183.6992 on 396 degrees of freedom

Number of Fisher scoring iterations: 4


Reference group is level  3  of the response
```

```
library(car)
Anova(modv)
```

```
Analysis of Deviance Table (Type II tests)

Response: Injury
         Df   Chisq Pr(>Chisq)
Clinic    4 11.9012     0.0181 *
Agegroup  2  0.0154     0.9923
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Odds Ratio - note that the confidence interval limits are reversed.
exp(-coef(modv))
```

```
(Intercept):1 (Intercept):2     Clinic2:1     Clinic2:2     Clinic3:1
```

```
    0.2135119     0.3497396     1.2449671     4.0763280     2.2747209
    Clinic3:2   AgegroupB:1   AgegroupB:2
    2.3623847     0.9545231     0.9818537
```

```
exp(-confint(modv))
```

```
                    2.5 %     97.5 %
(Intercept):1  0.4748214 0.09600943
(Intercept):2  0.8205713 0.14906415
Clinic2:1      3.1599718 0.49049272
Clinic2:2     12.1629723 1.36615043
Clinic3:1      6.9450478 0.74504244
Clinic3:2      7.9496566 0.70202549
AgegroupB:1    2.1274023 0.42827548
AgegroupB:2    2.5312568 0.38085301
```

3.8.3 Answer

There is a significant effect of clinic ($\chi^2 = 11.9, p = 0.02$), but no evidence of an age effect ($\chi^2 = 0.02, p = 0.99$).

The odds of receiving patients with face injuries in clinic 1 is 4.08 (1.37-12.2) times that of clinic 2. Meanwhile, the odds of receiving patients with finger injuries in clinic 1 is 1.25 (0.49-3.16) times that of clinic 2.

3.9 Exact Generalized Logits Test

- Exact methods are more appropriate for tables with small or zero counts.

Exact Method Warning

Generalized logits tests have more parameters and exact methods are typically more computationally expensive. When using exact methods, try to keep the model small for relatively small data sets.

Exact Method in SAS

The `exact` statement can perform joint tests as well as calculate exact confidence intervals for odds ratios. Use the "Exact" p-value column in the "Probability" row in the exact conditional tests.

3.10 Example

The following data come from a study in which a physical therapy practice experimented with different treatment modalities in a strengthening program. The therapists rotated patients through three programs for maintenance after basic strength had been attained. The strengthening programs included free weights, resistance bands, and weight machines. The therapists are interested in whether gender and age affect choice of strengthening programs.

3.10.1 Data Set

```
data pt;
format therapy $11.;
input gender $ age $ therapy $ count @@;
datalines;
males adult machines 2 males senior machines 10
males adult freeweights 13 males senior freeweights 9
males adult bands 3 males senior bands 3
females adult machines 3 females senior machines 8
females adult freeweights 9 females senior freeweights 0
females adult bands 1 females senior bands 1
;
```

3.11 Example

The following data come from a study in which a physical therapy practice experimented with different treatment modalities in a strengthening program. The therapists rotated patients through three programs for maintenance after basic strength had been attained. The strengthening programs included free weights, resistance bands, and weight machines. The therapists are interested in whether gender and age affect choice of strengthening programs.

> **Warning**
>
> Small cell counts means we need to use exact methods.

3.11.1 SAS Code

```
proc logistic data=pt;
title "asymptotic";
freq count;
```

```
class gender(ref='females') age(ref='senior') / param=ref;
model therapy(ref='machines') = gender age / link=glogit scale=none aggregate;
run;

proc logistic data=pt;
title "exact";
freq count;
class gender(ref='females') age(ref='senior') / param=ref;
model therapy(ref='machines') = gender age / link=glogit scale=none aggregate;
exact gender age/joint estimate=both;
run;
```

3.11.2 Answer

There is a marginal effect of gender ($ p=0.099), $but there is a significant effect of age$(^2 = 12.2, p=0.003$).

The odds of males preferring bands is 4.1 (0.49-59.55) times than that of females. Additionally, the odds of males preferring freeweights is 4.49 (0.93-30.2) times than that of females.

The odds of an adult preferring bands is 5.08 (0.60-53.2) times that of seniors. However, the odds of an adult preferring freeweights is 12.47 (2.7-84.0) times that of seniors.